OlomiPay

Building Trust Through Blockchain

Minimum Viable Product (MVP) Document

Mobile Money ↔ Stellar Payment Gateway & Wallet

Prepared for Investors & Regulators

Document v1.0 · 3 June 2026 · Status: Testnet MVP

CONFIDENTIAL

Table of Contents

1. Executive Summary

OlomiPay is a Progressive Web App (PWA) that bridges African mobile money (M-Pesa, Tigo Pesa, Airtel Money, MTN MoMo) with the Stellar blockchain, allowing users to hold value in USDC — a US-dollar-pegged stablecoin — instead of devaluing local currency.

Users deposit local currency, instantly receive USDC in a real Stellar wallet, and send money peer-to-peer, by phone number, by QR code, or inside a built-in encrypted chat. The platform earns a transparent 1% fee, collected automatically to a dedicated fee wallet.

A licensed liquidity provider (Yellow Card) handles local-currency ↔ USDC conversion, so OlomiPay never touches SWIFT and carries minimal float risk. The MVP is fully functional on Stellar Testnet with the Yellow Card sandbox, and is architected so the move to production requires configuration changes only — no code rewrite.

Product	OlomiPay — Mobile Money ↔ Stellar payment gateway & wallet
Status	Testnet MVP (Stellar Testnet + Yellow Card sandbox)
Backend / Frontend	v3.0.0 / v1.1.0
Settlement	Stellar (3–5 second finality, sub-cent network fees)
Stablecoin	USDC (Circle)
Markets	Tanzania (launch), Kenya, Uganda, +20 Yellow Card countries

2. Problem & Solution

The Problem

1. Currency devaluation — the Tanzanian Shilling and many African currencies lose value yearly; ordinary savings erode.
2. Expensive, slow transfers — cross-network and cross-border money movement is costly and slow.
3. Limited dollar access — ordinary users cannot easily hold USD; banks require SWIFT and high minimums.
4. Fragmented experience — money, chat, and bill payments live in separate apps.

The Solution

A single mobile-first wallet where users:

- Deposit mobile money and instantly receive USDC that holds its USD value.
- Send money to anyone in seconds — by phone, Stellar address, or QR code.
- Chat, and send or request money directly inside the conversation.
- Withdraw back to mobile money at any time.

All settlement is on Stellar: 3–5 second finality and fractions of a cent in network fees.

3. Target Users

Segment	Primary Need
Individuals (TZ / KE / UG)	Protect savings in USD, send to family, pay bills
Small merchants	Accept digital payments, QR checkout
Diaspora senders	Low-cost remittance into mobile money
Businesses	Payroll and bulk disbursement (Phase 2)

4. MVP Scope

In Scope — Built and Working

Core wallet & money movement

- Phone + 6-digit PIN registration; a real Stellar keypair is generated per user (secret encrypted with the PIN).
- USDC + XLM balances with live USD/TZS equivalents.
- Send USDC/XLM by Stellar address or phone number.
- QR receive (SEP-0007 URIs) and camera-based QR scanner for sending.
- Deposit via mobile money (STK Push) → Yellow Card conversion → USDC credited.
- Withdraw USDC → mobile money (B2C payout); full transaction history.

Fees & liquidity

- Transparent 1% platform fee, collected atomically on every transfer to a dedicated fee wallet.
- Full fee breakdown shown before confirming (mid-rate, provider spread, platform fee, network fee, net received).
- Yellow Card liquidity integration; sandbox mirrors mainnet fees exactly.

Encrypted chat & social payments

- 1-to-1 encrypted chat (NaCl keys), real-time delivery/read receipts, typing indicators.
- Send money inside chat with PIN authentication; payment requests with Accept / Decline.
- Global push + in-app sound notifications across all pages, with unread badges.

Admin & security

- Admin dashboard: users, transactions, fees, platform & fee wallets, PDF/CSV reports by date range.
- Role-based access control (Owner / Financial Controller / Developer / Viewer), enforced server-side.
- Maker-checker approval queue, TOTP 2FA, and step-up authentication for high-risk actions.
- Fail-open fraud gate on every send; auto-reconciler for stuck transactions; immutable audit log.

Out of Scope — Post-MVP

- Live mainnet money movement and production Yellow Card / Daraja credentials.
- Muxed/pooled custodial ledger (requires EMI/PSP license).
- FIDO2 hardware keys, USSD/SMS offline transactions, geographic HA cluster, ML fraud model.

5. How the Money Flows

Flow	Steps
Deposit	M-Pesa STK Push → Daraja callback → Yellow Card converts local→USDC → platform sends NET USDC to user (1% retained to fee wallet); network fee paid by platform.
Send (P2P)	User PIN → fraud gate (fail-open) → one atomic Stellar transaction: 99% to recipient + 1% to fee wallet.
Withdraw	User PIN → USDC pulled to platform → Yellow Card converts USDC→local → M-Pesa B2C payout to phone.

No SWIFT is involved. Conversion is handled by the licensed liquidity provider; OlomiPay holds a pre-funded USDC float.

6. Architecture & Technology

Resilience Built In

- Idempotent database migrations — no data loss on deploy.
- Atomic multi-operation Stellar transactions (all-or-nothing).
- Time bounds on every transaction — no stuck or replayed transactions.
- Auto-reconciler self-heals pending transactions; daily compliance checks toward a zero balance.

Technology Stack

Layer	Technology
Frontend	Next.js 14 (App Router), React 18, Tailwind CSS, PWA + Service Worker
Backend	Node 20, Express, TypeScript, Socket.io
Database	PostgreSQL via Prisma ORM (36 data models)
Blockchain	Stellar (Horizon + Soroban RPC), USDC asset, stellar-sdk v12
Smart contracts	Soroban (Rust): payments, bonds, chama, lending, savings, staking
Liquidity	Yellow Card Business API (local ↔ USDC)
Mobile money	Safaricom Daraja (STK Push + B2C)
Notifications	Web Push (VAPID) + in-app Web Audio
Hosting	Vercel (frontend) + Railway (backend + PostgreSQL)

7. Security & Compliance

Control	Implementation
PII off-chain	Names/phones in PostgreSQL; only hashes/memos on-chain
Encrypted secrets	Stellar secret keys encrypted with the user's PIN
RBAC / Zero-Trust	Every admin action checked server-side (prevents BFLA)
Step-up auth	Fresh TOTP required for high-risk admin actions
Fraud gate	Sub-second pre-flight screen on every transfer (fail-open)
Maker-checker	Two-person rule for sensitive money operations
Audit log	Immutable record of all back-office actions
Route protection	Middleware blocks unauthenticated access to the app

8. Success Metrics (MVP Validation)

Metric	Target
Testnet deposit → USDC credit completion	> 95%
P2P send confirmation time	< 5 seconds
Fee correctly collected to fee wallet	100% of transfers
Chat message delivery (online)	< 1 second
Fraud gate latency (rules tier)	< 50 ms
Data loss across deployments	0 (zero)
Users completing deposit → send → withdraw loop	≥ 20

9. Roadmap

Phase	Focus
Phase 1 — MVP (current)	Testnet wallet, deposit/withdraw, P2P, chat payments, admin, RBAC, fraud gate
Phase 2 — Production	Live Yellow Card + Daraja, mainnet USDC, float management, KYC tiers & limits, regulatory path
Phase 3 — Scale	Redis fraud store + async FinCrime agent, muxed ledger, daily auto-reconciliation, multi-region HA, passkeys, USSD/SMS
Phase 4 — Ecosystem	Merchant QR checkout, payroll/bulk disbursement, savings/staking/bonds, developer API

10. Key Risks & Mitigations

Risk	Mitigation
Liquidity / float shortfall	Provider model; daily reconciliation; float monitoring
Regulatory (custody / PSP)	Operate under partner license until Bank of Tanzania approval
Stablecoin de-peg	USDC (Circle) — most regulated stablecoin; monitor reserves
Fraud / money laundering	Fail-open gate now; ML model + sanctions screening in Phase 3
Rural connectivity	PWA offline shell now; USSD/SMS fallback in Phase 3
Key loss / recovery	PIN-encrypted keys; recovery flow hardened before production

11. Go / No-Go for Production

Before handling real money, the following are mandatory:

5. Production Yellow Card + Daraja credentials and signed agreements.
6. Legal: a PSP license or written partner-license coverage.
7. Separate, funded fee and float wallets on Stellar mainnet, with monitoring.
8. Rotation of all secrets and access tokens.
9. Penetration test of the admin panel and money-movement endpoints.
10. A KYC/AML provider integrated and transaction limits enforced.

This document reflects the actual implemented state of the OlomiPay repository as of 3 June 2026.